

Registration No: 2025/CS/079

Index No: 25000799

Exercise 01

01.

- | | |
|------------|------------|
| a) Invalid | b) Valid |
| c) Valid | d) Invalid |
| e) Invalid | f) Valid |
| g) Valid | h) Invalid |
| i) Invalid | j) Invalid |

02.

```
#include <stdio.h>
```

```
int main() {  
    printf("Size of int   : %zu\n", sizeof(int));  
    printf("Size of char  : %zu\n", sizeof(char));  
    printf("Size of long   : %zu\n", sizeof(long));  
    printf("Size of short  : %zu\n", sizeof(short));  
    printf("Size of float   : %zu\n", sizeof(float));  
    printf("Size of double   : %zu\n", sizeof(double));  
    printf("Size of long double : %zu\n", sizeof(long double));  
}
```

03.

(a)

```
# 1 "test11.c"
# 1 "<built-in>"
# 1 "<command-line>"
# 31 "<command-line>"
# 1 "/usr/include/stdc-predef.h" 1 3 4
# 32 "<command-line>" 2
# 1 "test11.c"
# 1 "/usr/include/stdio.h" 1 3 4
# 27 "/usr/include/stdio.h" 3 4
# 1 "/usr/include/aarch64-linux-gnu/bits/libc-header-start.h" 1 3 4
# 33 "/usr/include/aarch64-linux-gnu/bits/libc-header-start.h" 3 4
# 1 "/usr/include/features.h" 1 3 4
# 461 "/usr/include/features.h" 3 4
# 1 "/usr/include/aarch64-linux-gnu/sys/cdefs.h" 1 3 4
# 452 "/usr/include/aarch64-linux-gnu/sys/cdefs.h" 3 4
# 1 "/usr/include/aarch64-linux-gnu/bits/wordsize.h" 1 3 4
# 453 "/usr/include/aarch64-linux-gnu/sys/cdefs.h" 2 3 4
# 1 "/usr/include/aarch64-linux-gnu/bits/long-double.h" 1 3 4
# 454 "/usr/include/aarch64-linux-gnu/sys/cdefs.h" 2 3 4
# 462 "/usr/include/features.h" 2 3 4
# 485 "/usr/include/features.h" 3 4
# 1 "/usr/include/aarch64-linux-gnu/gnu/stubs.h" 1 3 4

# 1 "/usr/include/aarch64-linux-gnu/bits/wordsize.h" 1 3 4
# 6 "/usr/include/aarch64-linux-gnu/gnu/stubs.h" 2 3 4

# 1 "/usr/include/aarch64-linux-gnu/gnu/stubs-lp64.h" 1 3 4
# 9 "/usr/include/aarch64-linux-gnu/gnu/stubs.h" 2 3 4
# 486 "/usr/include/features.h" 2 3 4
# 34 "/usr/include/aarch64-linux-gnu/bits/libc-header-start.h" 2 3 4
# 28 "/usr/include/stdio.h" 2 3 4

extern int fileno_unlocked (FILE *__stream) __attribute__ ((__nothrow__ , __leaf__));
# 800 "/usr/include/stdio.h" 3 4
extern FILE *popen (const char *__command, const char *__modes);

extern int pclose (FILE *__stream);

extern char *ctermid (char *__s) __attribute__ ((__nothrow__ , __leaf__));
# 840 "/usr/include/stdio.h" 3 4
extern void flockfile (FILE *__stream) __attribute__ ((__nothrow__ , __leaf__));

extern int ftrylockfile (FILE *__stream) __attribute__ ((__nothrow__ , __leaf__));

extern void funlockfile (FILE *__stream) __attribute__ ((__nothrow__ , __leaf__));
# 858 "/usr/include/stdio.h" 3 4
extern int __uflow (FILE *);
extern int __overflow (FILE *, int);
# 873 "/usr/include/stdio.h" 3 4

# 2 "test11.c" 2

# 3 "test11.c"
int main() {
    printf("Hello World\n");
    return 0;
}
```

(b)

```
.arch armv8-a
.file "test11.c"
.text
.section .rodata
.align 3
.LC0:
.string "Hello World"
.text
.align 2
.global main
.type main, %function
main:
.LFB0:
.cfi_startproc
stp x29, x30, [sp, -16]!
.cfi_def_cfa_offset 16
.cfi_offset 29, -16
.cfi_offset 30, -8
mov x29, sp
adrp x0, .LC0
add x0, x0, :lo12:.LC0
bl puts
mov w0, 0
ldp x29, x30, [sp], 16
.cfi_restore 30
.cfi_restore 29
.cfi_def_cfa_offset 0
ret
.cfi_endproc
.LFE0:
.size main, .-main
.ident "GCC: (Ubuntu 9.4.0-1ubuntu1~20.04.2) 9.4.0"
.section .note.GNU-stack,"",@progbits
~
~
~
~
~
~
~
```

(c)

```
00000000: 7f45 4c46 0201 0100 0000 0000 0000 0000 .ELF.....
00000010: 0100 b700 0100 0000 0000 0000 0000 0000 .....
00000020: 0000 0000 0000 0000 0803 0000 0000 0000 .....
00000030: 0000 0000 4000 0000 0000 4000 0d00 0c00 ....@.....@....
00000040: fd7b bfa9 fd03 0091 0000 0090 0000 0091 .{.....
00000050: 0000 0094 0000 8052 fd7b c1a8 c003 5fd6 .....R.{...._
00000060: 4865 6c6c 6f20 576f 726c 6400 0047 4343 Hello World..GCC
00000070: 3a20 2855 6275 6e74 7520 392e 342e 302d : (Ubuntu 9.4.0-
00000080: 3175 6275 6e74 7531 7e32 302e 3034 2e32 1ubuntu1~20.04.2
00000090: 2920 392e 342e 3000 1000 0000 0000 0000 ) 9.4.0.....
000000a0: 017a 5200 0478 1e01 1b0c 1f00 2000 0000 .zR..x.....
000000b0: 1800 0000 0000 0000 2000 0000 0041 0e10 .....A..
000000c0: 9d02 9e01 46de dd0e 0000 0000 0000 0000 ....F.....
000000d0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000000e0: 0000 0000 0000 0000 0100 0000 0400 f1ff .....
000000f0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000100: 0000 0000 0300 0100 0000 0000 0000 0000 .....
00000110: 0000 0000 0000 0000 0000 0000 0300 0300 .....
00000120: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000130: 0000 0000 0300 0400 0000 0000 0000 0000 .....
00000140: 0000 0000 0000 0000 0000 0000 0300 0500 .....
00000150: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000160: 0a00 0000 0000 0500 0000 0000 0000 0000 .....
00000170: 0000 0000 0000 0000 0d00 0000 0000 0100 .....
00000180: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000190: 0000 0000 0300 0700 0000 0000 0000 0000 .....
000001a0: 0000 0000 0000 0000 0a00 0000 0000 0800 .....
000001b0: 1400 0000 0000 0000 0000 0000 0000 0000 .....
000001c0: 0000 0000 0300 0800 0000 0000 0000 0000 .....
000001d0: 0000 0000 0000 0000 0000 0000 0300 0600 .....
000001e0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000001f0: 1000 0000 1200 0100 0000 0000 0000 0000 .....
00000200: 2000 0000 0000 0000 1500 0000 1000 0000 .....
00000210: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000220: 0074 6573 7431 312e 6300 2464 0024 7800 .test11.c.$d.$x.
00000230: 6d61 696e 0070 7574 7300 0000 0000 0000 main.puts.....
00000240: 0800 0000 0000 0000 1301 0000 0500 0000 .....
00000250: 0000 0000 0000 0000 0c00 0000 0000 0000 .....
00000260: 1501 0000 0500 0000 0000 0000 0000 0000 .....
```

(d)

```
#include <stdio.h>

int main() {
    // this is a comment
    printf("Hello World\n");
    return 0;
    // here's another one.
}
```

```
# 3 "test11.c"
int main() {

    printf("Hello World\n");
    return 0;

}
```

(e)

```
.LC0:
    .string "Hello World"
    .text
    .align 2
    .global main
    .type main, %function
main:
.LFB0:
```

04.

```
#include <stdio.h>
#include <limits.h>
#include <float.h>

int main() {
    printf("Range of Int : %d to %d\n", INT_MIN, INT_MAX);
    printf("Range of char : %d to %d\n", CHAR_MIN, CHAR_MAX);
    printf("Range of long : %ld to %ld\n", LONG_MIN, LONG_MAX);
    printf("Range of short : %d to %d\n", SHRT_MIN, SHRT_MAX);
    printf("Range of float : %f to %f\n", FLT_MIN, FLT_MAX);
    printf("Range of double : %f to %f\n", DBL_MIN, DBL_MAX);
    printf("Range of long double : %Lf to %Lf\n", LDBL_MIN,
        LDBL_MAX);
}
```

05.

(a)

```
#include <stdio.h>

int main() {
    double x = 125.456734516547332453425;
    printf("i)    %.6f\n", x);
    printf("ii)   %.3f\n", x);
    printf("iii)  %+ .0e\n", x);
    printf("iv)   %.6E\n", x);
    printf("v)    %A\n", x);
}
```

(b)

```
#include <stdio.h>
```

```
int main() {
    int y = 76;
    printf("i)    %d\n", y);
    printf("ii)   %x\n", y);
    printf("iii)  %X\n", y);
    printf("iv)   %o\n", y);
    printf("v)    %#x\n", y);
    printf("vi)   %#X\n", y);
    printf("vii)  %#o\n", y);
    printf("viii) %010d\n", y);
    printf("ix)   %10d\n", y);
}
```

(c)

```
#include <stdio.h>
```

```
int main() {
    int day, month, year, hour, minute;
    day = 30;
    month = 4;
    year = 2024;
    hour = 12;
    minute = 57;

    printf("i)    %d-%02d-%d %d:%d\n", year, month, day, hour,
minute);
    printf("ii)   %d/%02d/%d %d:%d Hrs\n", day, month, year, hour,
minute);
    printf("iii)  %d.%d.%d %d Hours %d Minutes\n", month, day,
year, hour, minute);
    printf("iv)   %x%d%d-%03d%03d \n", year, month, day, hour,
minute);
}
```

(d)

```
#include <stdio.h>
```

```
int main() {  
    char c = 'A';  
    printf("i)  %c\n", c);  
    printf("ii) %d\n", c);  
}
```